

9	A car of mass 1200 kg travels along a horizontal road at a speed of 10 m s⁻¹. At the time it just begins to accelerate at 0.20 m s⁻², the total resistive force acting on the car is 160 N. What is the total output power developed by the car as it just begins its acceleration?			
	A 800 W	B 1600 W	C 2400 W	D 4000 W
L2	Answer: D Apply Newton's second law, $F_D - F_R = ma$ $F_D - 160 = (1200)(0.20)$ $F_D = 400 \text{ N}$ Output power = $F_D \times v = 400 \times 10 = \mathbf{4000 \text{ W}}$ <i>F_D is the driving force</i> <i>F_R is the total resistive force</i>			

10 The diagram shows a projectile being launched at an angle θ to the vertical.